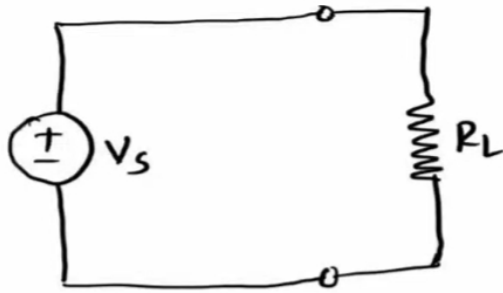


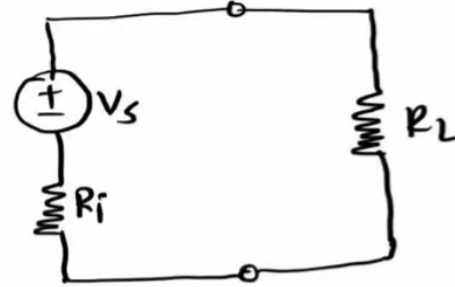
# Ideal v/s. Practical Voltage Source :-



Ideal Voltage Source

→ It has zero internal resistance

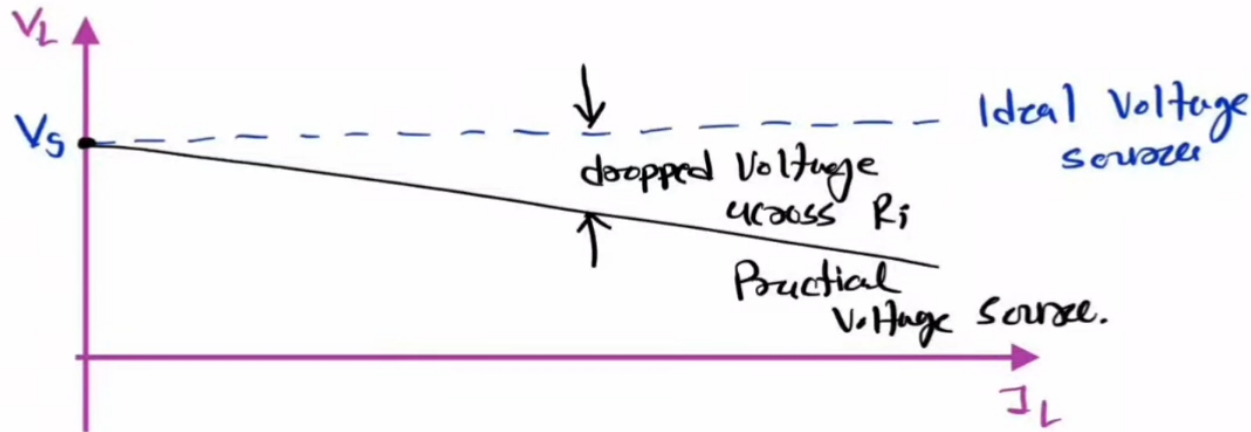
→ It delivers constant voltage across load  $R_L$ .



Practical Voltage Source.

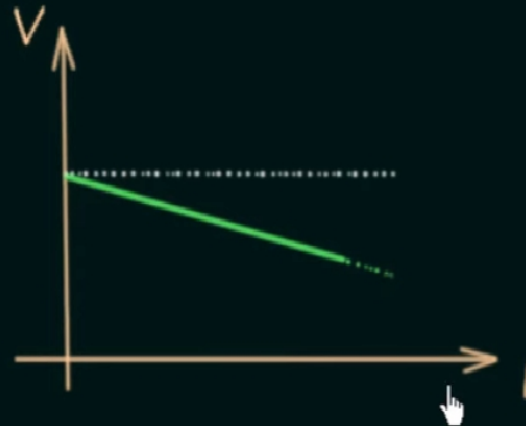
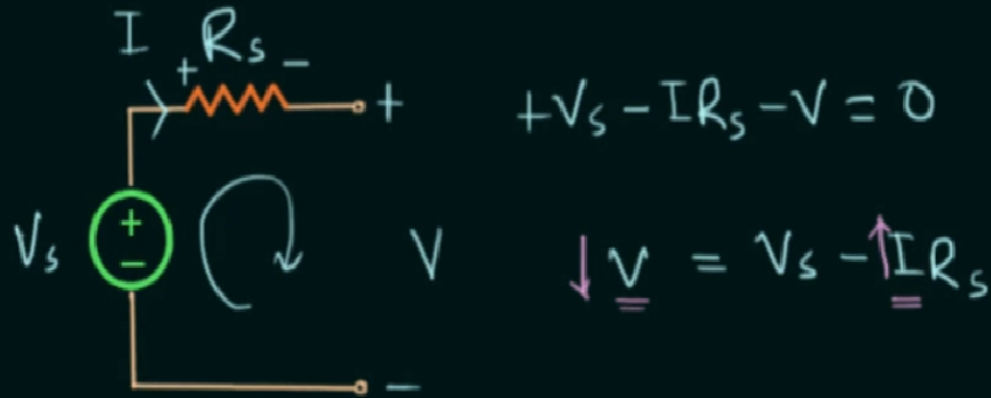
→ It has  $R_i$  internal resistance connected in series with voltage source.

→ It has some voltage drop across  $R_i$ , so can not deliver  $V_s$  across load.



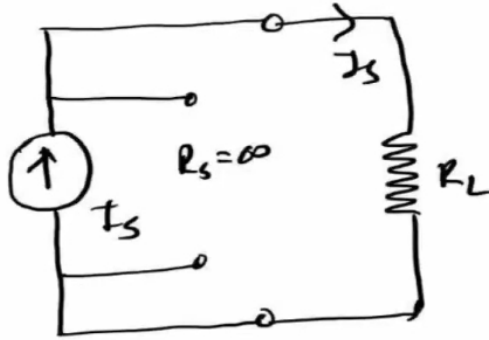
$$R_s \neq 0$$

**Practical Voltage Source:** has a finite internal resistance and it delivers the energy at a specified voltage, which depends on the current delivered by the source.

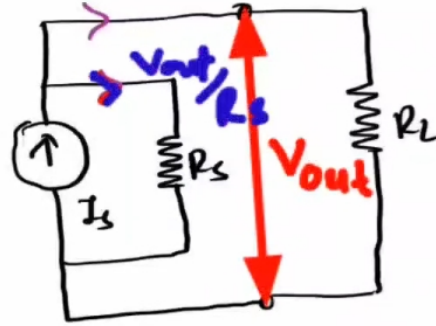


As we draw more current( $I$ ) from the source  $V_s$ , the voltage drop across resistor( $R_s$ ) increases and resultant Voltage( $V$ ) across terminals reduces.

# Ideal v/s. Practical Current Source :-



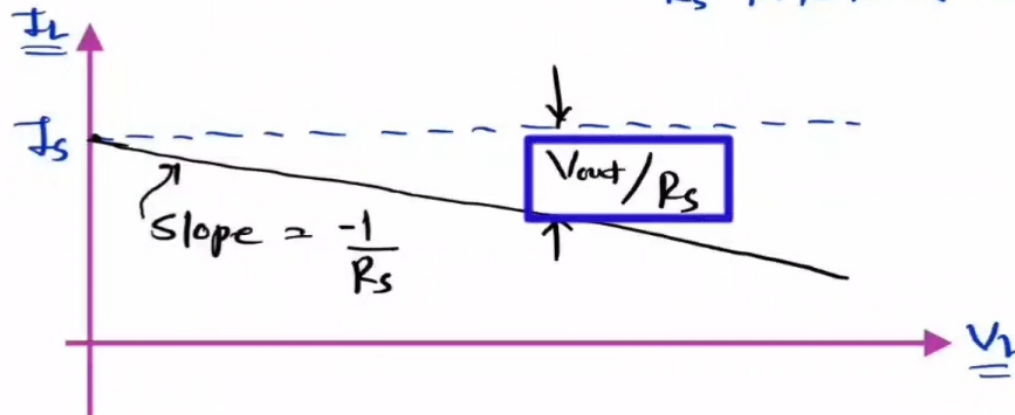
Ideal Current Source



Practical Current Source

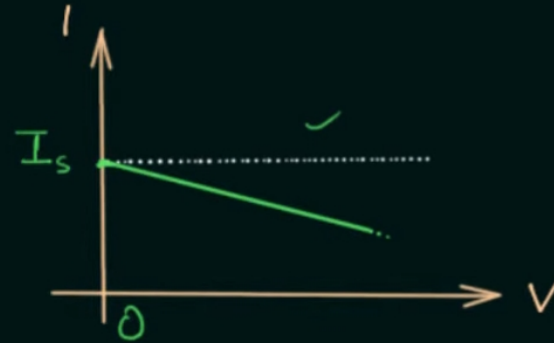
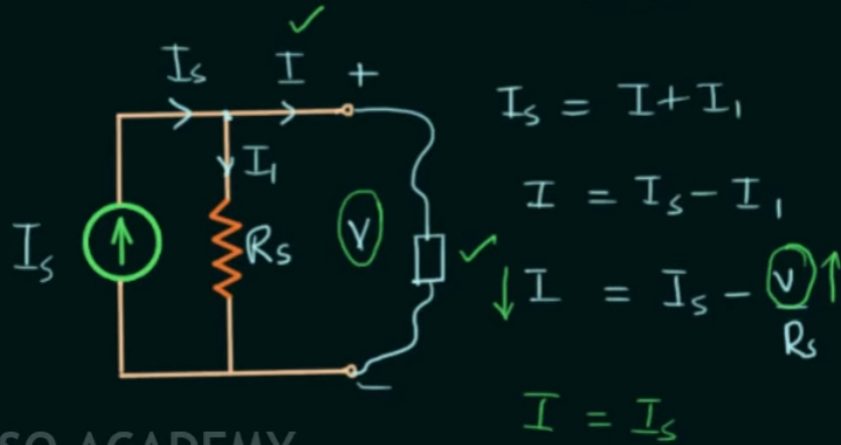
- It has infinite resistance in parallel across current source  $I_s$
- It delivers constant current to load  $R_L$

- It has finite resistance  $R_s$  connected in parallel to current source  $I_s$ .
- It can not give constant current to load as fractional current goes through  $R_s$  internal resistance.



$$R_s \neq \infty$$

**Practical Current Source:** has finite internal resistance and it delivers the energy at a specified current which is dependent on the voltage across the source.



NESO ACADEMY

As we draw more voltage(V) across the load, Current through the source ( $I_s$ ) decreases.